

Agricultural Intensification and Extensification



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Intensification

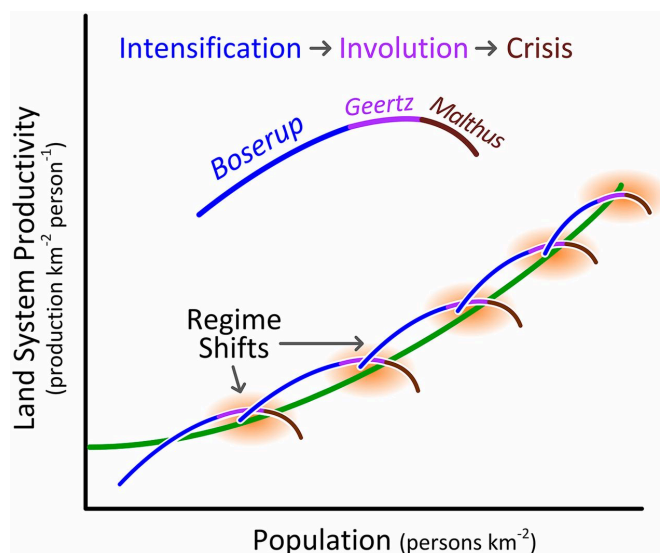
Agricultural intensification is the process by which people increase food yields through additional labor inputs. This typically results in higher yields of crops or animal products per unit of land area. A variety of strategies can be used to achieve these aims. Farmers can plant multiple crops in a single field, either together or in sequence, with one planted following the harvest of the other. Such systems are termed multicropping or **polycropping** strategies. Farmers can also raise yields by reducing weed competitors in fields through weeding and plowing, or by increasing the growth rate of plants through **manuring** or **irrigating** fields. Pastoralists can intensify by raising animals that eat different types of forage in an area, allowing more animals to feed per unit of land, or by making more efficient use of animal resources. For example, pastoral societies of East Africa, such as the Maasai, rely on their animals for multiple forms of food: milk and meat, but also blood, small amounts of which are regularly drained from cattle and cooked as a renewable food resource.

Intensive agricultural systems are those in which increases in labor inputs result in increased food yields per unit of area.

Note that technological innovation is important for many forms of agricultural intensification. Metal plows, irrigation pumps, chemical fertilizers, and nonhuman labor (from animals, such as oxen or donkeys, or from tractors) have all led to substantial increases in crop yields per unit of area. Such practices enable previously noncultivable lands to be farmed, increasing the area from which crops are produced and, thus, increasing overall agricultural productivity.

Population increase is often seen as the driving force behind the need to intensify agriculture. If you have enough food, why invest more time and labor in agriculture? Scholars have long argued about the relationship between population growth and food supply; most famously, Thomas Malthus, a theorist of the 18th and 19th centuries, argued that human population growth would outstrip our ability to produce enough food, leading to famine and misery. Esther Boserup, an

economist of the 20th century, observed an opposite trend: Technological innovation and ingenuity lead to increased crop yields as needed, allowing a given unit of area to support greater populations over time, albeit with increased labor inputs. The anthropologist Clifford Geertz published an influential book in 1963 ([BU Library version here \(https://www.jstor.org.ezproxy.bu.edu/stable/jj.8501191\)](https://www.jstor.org.ezproxy.bu.edu/stable/jj.8501191)) in which he describes the observation that, as populations increase, increases in agricultural efficiency (as noted by Boserup) alternate with periods in which increased labor is needed just to maintain yields, resulting in an agricultural system that may fail to maintain sufficient production for the increased population. These periods of stagnation are termed “**involution**” by Geertz. The relationship among these three “phases” of agricultural innovation, and their relationship with population growth, is depicted in the figure below:



This figure—from the article “Used Planet: A Global History” by Erle Ellis et al. (2013)—depicts the conceptual relationship among the phases of agricultural innovation defined by Boserup, Geertz, and Malthus during agricultural intensification.

Source: Ellis, E. C., Kaplan, J. O., Fuller, D. Q., Vavrus, S., Klein Goldewijk, K., & Verburg, P. H. (2013). Used planet: A global history. *Proceedings of the National Academy of Sciences*, 110(20), 7978–7985. <https://doi.org/10.1073/pnas.1217241110>

Agricultural **involution** is a period in which increased labor is needed to maintain crop yields, without increasing yields per unit of area or overall.

Extensification

Extensive agriculture, in contrast, typically involves large areas of land with low labor inputs. While the per-unit-area yield is lower than that of intensive agricultural systems, labor inputs are also much lower, so extensive agriculture is often more efficient in terms of crop yield per unit of labor. Extensive agricultural strategies include **fallowing** land, or allowing it one or more growing seasons without cultivation, which can restore soil nutrients absorbed by hungry crops. A similar strategy is swidden cultivation, also known as slash-and-burn cultivation. In swidden cultivation, natural ecological cycles are mimicked by farmers, who cut dense, woody vegetation, burn it to release its nutrients into the soil, plant for one or a few years in the nutrient-rich soil, and then move on to another plot of regrown forest to repeat the cycle, allowing the last piece of land to fallow and regrow woody vegetation. This can be a sustainable strategy that allows natural plant succession to rebuild soil nutrition and minimizes the labor needed to produce good crop yields. Extensive pastoral strategies rely on the mobility of animals, and sometimes of the human communities that depend on them, to access seasonally available resources and prevent overgrazing, which can lead to long-term reduction in the forage value of an area.

Extensive agricultural systems are those that reduce the amount of labor needed per unit of food yield by increasing the area of land under cultivation or pastoral production.

Why Intensify or Extensify Agriculture?

How do societies choose one or another strategy to respond to food shortages or opportunities to produce a food surplus? How do individual farmers and herders operate within societies when they may have different economic needs, resources available, or risk tolerances? These are questions that archaeologists (and scholars of historical and contemporary agriculture) continue to ask, as understanding human decision-making, individual or collective, is never simple. More than a century of research on such topics, however, indicates that several variables are important.

Human population density has always been a major driver of subsistence decisions. If a family is hungry, they will take steps to increase their food supply. This may involve increasing their labor inputs, bringing new land into cultivation, or investing in durable agricultural technologies (such as new tools, draft animals, or irrigation systems) that can increase yields. Without question, human societies respond to population increases through a variety of strategies to increase food supply.

More complex is understanding how both individuals and communities assess and respond to agricultural risk. There are myriad environmental risks for farmers and herders: droughts, floods, insect pests, diseases, and conflict with other people (e.g., cattle raiding, burning an enemy's crops in the fields). They address these risks through a variety of flexible strategies that they employ based on local circumstances and particular risks faced. For example, in dry areas, drought may be a major concern, but droughts (even when common) rarely strike for more than two consecutive years. If it is possible to store three years' worth of surplus grain, then it is very unlikely that

drought will cause starvation. But this also depends on having crops that can be stored for that long and storage systems that will keep the food safe from pests and decay. We delve deeper into the complexity of food-storage systems in a future module.

Agricultural strategies, however, can also mitigate risk. For example, hail or flood might destroy crops in a field, but if a family farms five different fields, located in different areas of the landscape (e.g., in the floodplain, on a hill, protected by trees), then it is much less likely that any single hazard will destroy all of their food supply. Predictable risks, such as drought, can be ameliorated through irrigation systems that deliver water flows that are more predictable than rainfall. Raising both animals and plants is often an effective strategy to reduce a variety of risks. For example, if the neighboring society invades your land, you can take your herds and leave. Even if your fields are burned, you have animal resources available until you can replant. Farming multiple crops can minimize the effects of disease or insect damage, and drought-resistant crops produce more reliable yields in dry years.

Importantly, however, many agricultural practices and innovations also lead to additional risks. If drought risk is reduced through the installation of irrigation systems, but severe drought or changes in river dynamics reduce water flows to those irrigation canals, farmers may face catastrophic crop failures. This is one reason why multiple methods of risk management are often employed together, with each serving as a backup plan for the unforeseen failure of another. A greater threat to agricultural systems is increasing reliance on a complex system of technologies that cannot be easily substituted. For example, modern industrialized agriculture is greatly dependent on the use of chemical fertilizers, pesticides and herbicides, and fossil-fuel-powered tractors to produce the high yields on which we depend. Severe disruptions to international trade in fertilizers or oil can threaten the entire system. Can you imagine farmers in the U.S. Midwest returning to ox-pulled plows to farm hundreds of acres? Such complex agricultural systems create **path dependencies**, systems in which decisions made in the past (such as investing in tractors and then expanding farmlands to make efficient use of those tractors) limit the set of options possible in the present and near future (it is impossible to farm such large areas with animal-driven plows). Path dependencies exist in a variety of human technological systems, not just agriculture, but due to the centrality of agriculture to human life, such path dependencies are particularly important in the history of agricultural development.

Path dependencies result from past decisions that limit the range of present and future options. For example, agriculture itself created a path dependency: It would be impossible to feed the world's current population through hunting and gathering. We don't have an alternative to agriculture if we are to feed the world's current population.